



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- i. ✓ What are basic mechanism of mass transport in voltammetry?
- ii. ✓ Differentiate between diffusion current and migration current.
- iii. ✓ Briefly explain cell constant.
- iv. ✓ What is role of convection on Faradaic current in Voltametric measurement?
- v. ✓ Define equivalent conductance.
- vi. ✓ Describe two disadvantages of amperometry.
- vii. ✓ What change in conductance when weak acid reacts with strong base?
- viii. ✓ Why mercury electrode is widely used?
- ix. ✓ What is electrical spark?
- x. ✓ Discuss characteristics of Arc sources.
- xi. ✓ How potentiometric titration is useful for determination of indicator for endpoint of titration?
- xii. ✓ Discuss programmable automation.
- xiii. ✓ What is voltammogram?
- xiv. ✓ How Voltametric techniques are classified?
- xv. ✓ Define migration current.

Answer the following questions.

(3x10=30)

- Q.2. (a) ✓ Explain in detail anodic stripping voltammetry.
(b) ✓ Write down application of voltammetry.
- Q.3. (a) ✓ Explain in detail four types of potentiometric titration.
(b) ✓ Describe applications, advantages and disadvantages of amperometer.
- Q.4. (a) ✓ Describe the instrumentation of electric arc/spark.
(b) ✓ Explain half wave potential with suitable example.

Analytical sp.1

Chem 431

SPRING 2024

SHORT QUESTIONS

Q1.explain mechanism of mass transfer in voltammetry.

ANS.*In voltammetry, mass transport refers to the movement of analyte molecules or ions from the bulk solution to the electrode surface. The basic mechanisms of mass transport in voltammetry include:*

1. Diffusion

Diffusion is the movement of analyte molecules or ions from a region of high concentration to a region of low concentration. In voltammetry, diffusion occurs when the analyte is consumed or produced at the electrode surface, creating a concentration gradient.

2. Migration

Migration is the movement of charged analyte molecules or ions in response to an electric field. In voltammetry, migration occurs when the electrode is charged, and the electric field influences the movement of ions.

3. Convection

Convection is the movement of analyte molecules or ions due to hydrodynamic forces, such as stirring or flow. In voltammetry, convection can enhance mass transport by bringing fresh analyte to the electrode surface.

Q 2..

Differentiate between diffusion current and migration current.

ANS:*Diffusion Current*

Diffusion current is the current generated by the movement of analyte molecules or ions from a region of high concentration to a region of low concentration due to diffusion. This type of current is driven by a concentration gradient.

- *Caused by*: Concentration gradient

- ***Direction*:** From high concentration to low concentration
- ***Dependent on*:** Concentration gradient, diffusion coefficient, and electrode surface area

Migration Current

Migration current is the current generated by the movement of charged analyte molecules or ions in response to an electric field. This type of current is driven by the electrostatic attraction or repulsion between the charged species and the electrode.

- ***Caused by*:** Electric field
- ***Direction*:** Towards or away from the electrode, depending on the charge of the species
- ***Dependent on*:** Electric field strength, charge of the species, and mobility of the species.

Q3. Explain cell constant.

Ans; The cell constant is a parameter used in electrochemistry to characterize the geometry of an electrochemical cell. It's defined as the ratio of the distance between the electrodes (l) to the surface area of the electrodes (A).

Cell Constant Formula

$$\text{Cell Constant (G}^*) = l / A$$

Where:

- l = distance between electrodes
- A = surface area of electrodes

Importance of Cell Constant

The cell constant is important because it:

- ***Affects Electrochemical Measurements*:** The cell constant influences the electrochemical response, including the current and potential measurements.
- ***Characterizes Cell Geometry*:** The cell constant provides information about the geometry of the electrochemical cell, which can impact the accuracy and reliability of measurements.

Q.4. what is role of convection on Faradaic current in voltammetric measurements?

ANS;

In voltammetric measurements, convection can significantly impact the current response.

Role of Convection

Convection refers to the movement of fluid (electrolyte solution) due to external forces, such as stirring or flow. In voltammetry, convection can:

- ***Enhance Mass Transport***: Convection can increase the rate of mass transport of analyte molecules or ions to the electrode surface, leading to increased current.
- ***Increase Current Response***: By bringing fresh analyte to the electrode surface, convection can increase the current response, especially at higher flow rates or stirring speeds.
- ***Improve Sensitivity***: Convection can improve the sensitivity of voltammetric measurements by enhancing the mass transport and increasing the current response.

Q.5.

Define equivalent conductance.

ANS: Equivalent conductance is a measure of the conductivity of an electrolyte solution. It's defined as the conductance of a solution containing one equivalent of an electrolyte when placed between two parallel electrodes having a unit length and large distance apart.

Equivalent Conductance Formula

$$\text{Equivalent Conductance } (\Lambda) = \kappa \times V$$

Where:

- κ = specific conductance (conductivity) of the solution
- V = volume containing one equivalent of an electrolyte
- ***Unit***: Equivalent conductance is typically expressed in units of $S \text{ cm}^2/\text{equiv}$ (Siemens per equivalent).

Q.6. write Two disadvantages of amperometry.

Ans: Here are two disadvantages of amperometry:

- **Interference from Other Electroactive Species**

Amperometry can be affected by the presence of other electroactive species in the solution, which can oxidize or reduce at the same potential as the analyte. This can lead to interference and inaccurate measurements.

- **Fouling of the Electrode**

Amperometry can be affected by fouling of the electrode surface, which can occur due to the adsorption of species onto the electrode surface. This can lead to a decrease in the electrode's sensitivity and accuracy over time, requiring frequent cleaning or replacement of the electrode.

These limitations highlight the importance of careful experimental design and electrode maintenance in amperometric measurements.

Q.7. what change in conductance when weak acid react with strong base?

ANS;

Reaction with Strong Base

When a strong base is added to the weak acid, a neutralization reaction occurs, forming a salt and water. The weak acid is converted into its conjugate base, which is a stronger electrolyte.

Changes in Conductance

As the reaction proceeds, the conductance of the solution:

1. ***Increases***: Due to the formation of more ions (conjugate base and cations) from the weak acid.
2. ***Becomes more conductive***: The solution becomes more conductive as the strong base reacts with the weak acid, producing more ions.

Q.8.

Why mercury is widely used as an electrode?

ANS:

Mercury is used as a working electrode, because of its atomic smooth surface and ability to act as an ideal polarized electrode.

Bcz it is a liquid and can be restored After each droplet.

It has high over potential for hydrogen evolution.

Mercury has high sensitivity and selectivity so it is widely used as an electrode.

Q.9 what's an electrical spark?

ANS: An electrical spark is a sudden, brief release of electrical energy that creates a visible flash of light and often produces a sharp cracking or popping sound. It occurs when there is a significant voltage difference between two points, causing the air or another medium to break down and conduct electricity.

Characteristics of Electrical Sparks

- ***High Voltage*:** Electrical sparks require a high voltage to initiate the breakdown of the medium.
- ***Brief Duration*:** Sparks are typically very short-lived, lasting only a fraction of a second.
- ***Visible Light*:** The spark produces a visible flash of light, which can be intense and bright.
- ***Audible Sound*:** The spark often produces a sharp cracking or popping sound.

Q.10. Discuss characteristics of arc sources.

Ans: Key Characteristics

- ***High Current*:** Arc sources operate at high currents, often in the range of tens to thousands of amperes.
- ***Low Voltage*:** Arc sources typically operate at relatively low voltages, often in the range of tens to hundreds of volts.
- ***High Temperature*:** The arc plasma can reach extremely high temperatures, often exceeding 5,000°C to 20,000°C or more.
- ***High Energy Density*:** Arc sources can deliver high energy densities, making them suitable for applications such as welding and cutting.
- ***Sustained Discharge*:** Unlike sparks, arc sources produce a sustained electrical discharge, which can be stable or unstable depending on the application.

Q.11. How potentiometric titration is useful for determination of indicator for end point?

Ans: In potentiometric titration, the potential difference between an indicator electrode and a reference electrode is measured as a function of the volume of titrant added. The endpoint is determined by monitoring the change in potential, which corresponds to the equivalence point of the reaction.

- **Determining Indicator for Endpoint**

To determine the indicator for the endpoint, potentiometric titration can be used to:

1. ***Identify the Equivalence Point***: By monitoring the potential change, the equivalence point can be identified, and the corresponding volume of titrant can be determined.
2. ***Select a Suitable Indicator***: Based on the potentiometric titration curve, a suitable indicator can be selected that changes color at or near the equivalence point.

Q.12

Discuss programmable automation.

Ans: Programmable automation refers to the use of control systems, such as computers or programmable logic controllers (PLCs), to automate industrial processes or machines. Here's a discussion on programmable automation:

Characteristics

1. ***Flexibility***: Programmable automation allows for flexibility in production, enabling changes to be made quickly and easily.
2. ***Programmability***: The control system can be programmed to perform specific tasks, and the program can be modified as needed.
3. ***Automation***: Programmable automation reduces the need for human intervention, increasing efficiency and productivity.

Q.13

What's voltammogram?

ANS: A voltammogram is a graphical representation of the current-voltage relationship in an electrochemical cell. It's a plot of the current (I) versus the potential (E) applied to the working electrode.

Characteristics

- ***Current-Potential Relationship***: A voltammogram shows the relationship between the current and potential, providing information about the electrochemical reaction.
- ***Peak-Shaped Curve***: Voltammograms often exhibit peak-shaped curves, which correspond to the oxidation or reduction of specific species.
- ***Scan Rate Dependence***: The shape and position of the voltammogram can depend on the scan rate, which is the rate at which the potential is changed.

Q.14. How voltammetric techniques are ?

ANS;

There are a variety of voltammetric methods.

anodic stripping voltammetry (ASV),

linear sweep voltammetry

, and cyclic voltammetry (CV).

Voltammetric methods typically involve the use of microelectrodes that frequently have areas on the order of $0.3\text{-}10\text{ cm}^2$.

Q.15. Define migration current

Ans: Migration current is a type of current that arises from the movement of ions in an electric field. In electrochemistry, migration current occurs when ions in the solution migrate towards the electrode due to the applied electric field.

Characteristics

- ***Electric Field-Driven***: Migration current is driven by the electric field applied to the solution.
- ***Ion Movement***: The current is generated by the movement of ions towards the electrode.
- ***Contribution to Total Current***: Migration current contributes to the total current measured in an electrochemical experiment.

Factors Affecting Migration Current

- ***Ion Concentration***: The concentration of ions in the solution affects the magnitude of the migration current.
- ***Electric Field Strength***: The strength of the electric field applied to the solution influences the migration current.
- ***Ion Mobility***: The mobility of ions in the solution affects their ability to migrate towards the electrode.

LONG QUESTIONS

Q. 2

Part (a)

Explain anodic stripping voltammetry

ANS;

Anodic stripping voltammetry (ASV) is a sensitive electroanalytical technique used to detect and quantify metal ions in solution.

Principle

1. ***Preconcentration***: Metal ions are preconcentrated onto a working electrode, typically a mercury film electrode or a solid electrode, by reducing them to their elemental form.
2. ***Stripping***: The preconcentrated metal is then oxidized (stripped) from the electrode by applying a positive potential scan.
3. ***Current Measurement***: The current generated during the stripping process is measured, which is proportional to the concentration of metal ions in the solution.

Steps

1. ***Deposition***: Metal ions are reduced and deposited onto the working electrode at a negative potential.
2. ***Equilibration***: The electrode is held at the deposition potential for a short period to allow equilibration.
3. ***Stripping***: The potential is scanned in the positive direction, causing the deposited metal to be oxidized and stripped from the electrode.

Advantages

1. ***High Sensitivity***: ASV is highly sensitive and can detect metal ions at very low concentrations.
2. ***Multi-Element Analysis***: ASV can be used to detect multiple metal ions simultaneously.
3. ***Low Detection Limits***: ASV can achieve low detection limits, making it suitable for trace analysis.

Applications

1. ***Environmental Monitoring***: ASV is used to detect and quantify metal ions in environmental samples, such as water and soil.
2. ***Biological Samples***: ASV can be used to analyze biological samples, such as blood and urine, for metal ion content.
3. ***Industrial Applications***: ASV is used in various industrial applications, including quality control and process monitoring.

By using ASV, researchers can accurately detect and quantify metal ions in various samples, making it a valuable tool in analytical chemistry..

Part (b)

Write down applications of VOLTAMMETRY.

ANS;

Voltammetry has a wide range of applications in various fields, including:

1. Environmental Monitoring

- ***Water Quality Analysis*:** Voltammetry is used to detect and quantify pollutants, such as heavy metals, in water samples.
- ***Soil Analysis*:** Voltammetry can be used to analyze soil samples for contaminants, such as heavy metals.

2. Biomedical Research

- ***Biomarker Detection*:** Voltammetry is used to detect biomarkers for diseases, such as cancer and neurodegenerative disorders.
- ***Neurotransmitter Analysis*:** Voltammetry can be used to study neurotransmitters and their role in various neurological disorders.

3. Pharmaceutical Analysis

- ***Drug Development*:** Voltammetry is used to study the electrochemical properties of pharmaceuticals and their metabolites.
- ***Quality Control*:** Voltammetry can be used to analyze pharmaceutical products for impurities and degradation products.

4. Food and Beverage Analysis

- ***Food Safety*:** Voltammetry is used to detect contaminants, such as pesticides and heavy metals, in food samples.
- ***Beverage Analysis*:** Voltammetry can be used to analyze beverages for antioxidants, flavor compounds, and other constituents.

5. Industrial Applications

- ***Process Monitoring*:** Voltammetry is used to monitor industrial processes, such as electroplating and corrosion.
- ***Quality Control*:** Voltammetry can be used to analyze industrial products for impurities and defects.

6. Forensic Analysis

- ***Toxicology*:** Voltammetry is used to detect and quantify toxic substances, such as drugs and poisons, in forensic samples.

Ques#3

Part (a)

Explain four types of potentiometric titration in details.

ANS;

Potentiometric titration is a technique used to determine the concentration of a substance by measuring the potential difference between two electrodes. Here are four types of potentiometric titration:

1. Acid-Base Titration

- ***Principle*:** Acid-base titration involves the reaction between an acid and a base to form a salt and water.
- ***Electrode*:** A glass electrode or a pH electrode is used to measure the potential difference.
- ***Application*:** Acid-base titration is used to determine the concentration of acids or bases in a solution.

2. Redox Titration

- ***Principle*:** Redox titration involves the transfer of electrons between two species, resulting in a change in oxidation state.
- ***Electrode*:** A platinum electrode or a gold electrode is used to measure the potential difference.
- ***Application*:** Redox titration is used to determine the concentration of oxidizing or reducing agents in a solution.

3. Precipitation Titration

- ***Principle*:** Precipitation titration involves the formation of a precipitate between the analyte and the titrant.



3. Precipitation Titration

- ***Principle***: Precipitation titration involves the formation of a precipitate between the analyte and the titrant.
- ***Electrode***: A silver electrode or a mercury electrode is used to measure the potential difference.
- ***Application***: Precipitation titration is used to determine the concentration of ions, such as chloride or bromide, in a solution.

4. Complexometric Titration

- ***Principle***: Complexometric titration involves the formation of a complex between the analyte and the titrant.
- ***Electrode***: A mercury electrode or a metal-specific electrode is used to measure the potential difference.
- ***Application***: Complexometric titration is used to determine the concentration of metal ions, such as calcium or copper, in a solution ..

Part (b)

Explain applications, advantages & disadvantages of amperometer.

ANS;

Amperometers (ammeters) have various applications:

1. ***Electrical Troubleshooting***: Diagnose electrical issues in circuits and systems.
2. ***Industrial Automation***: Monitor current consumption of machinery and equipment.
3. ***Power Systems***: Measure current in power distribution systems and detect overloads.
4. ***Automotive***: Diagnose electrical system issues in vehicles.
5. ***Renewable Energy***: Monitor current output of solar panels and wind turbines.
6. ***Research and Development***: Measure current in experimental circuits and devices.
7. ***Safety and Protection***: Detect overcurrent conditions to prevent damage or fire.

These applications highlight the importance of amperometers in ensuring efficient and safe operation of electrical systems. Here are some disadvantages of amperometers:

1. Insertion Loss

- Amperometers can introduce resistance into the circuit, affecting measurement accuracy.

2. Limited Range

- Amperometers may have limited current measurement ranges, requiring multiple devices for different applications.

3. Interference

- External magnetic fields or electrical noise can interfere with amperometer measurements.

4. Calibration Issues

- Amperometers require regular calibration to ensure accuracy, which can be time-consuming and costly.

5. Safety Risks

- Improper use of amperometers can pose electrical shock hazards or damage equipment.

6. Measurement Errors

- Amperometers can be affected by factors like temperature, humidity, or circuit conditions, leading to measurement errors.

7. Invasive Measurement

- Some amperometers require breaking into the circuit to insert the device, which can be inconvenient or disrupt the system.

These disadvantages highlight the need for careful selection, calibration, and use of amperometers to ensure accurate and reliable measurements. Here are some advantages of amperometers:

1. Accurate Current Measurement

- Amperometers provide precise measurements of electrical current, helping diagnose and troubleshoot electrical issues.



- Amperometers provide precise measurements of electrical current, helping diagnose and troubleshoot electrical issues.

2. Real-Time Monitoring

- Amperometers enable real-time monitoring of current flow, allowing for prompt detection of anomalies or faults.

3. Wide Range of Applications

- Amperometers are used in various fields, including electrical engineering, industrial automation, automotive, and renewable energy.

4. High Sensitivity

- Amperometers can detect small changes in current, making them useful for precise measurements.

5. Troubleshooting and Debugging

- Amperometers help identify electrical issues, such as shorts, overloads, or faulty components.

6. Preventive Maintenance

- Amperometers can detect potential problems before they cause equipment damage or downtime.

7. Improved Safety

- Amperometers help ensure safe operation of electrical systems by detecting overcurrent conditions or faults.

These advantages highlight the importance of amperometers in ensuring efficient, safe, and reliable operation of electrical systems.

Q no..4

Part (a)

Describe instrumentations of electric arc.

Ans:

Electrical arc instrumentation involves measuring and monitoring the electrical arc phenomenon in various applications, such as electrical power systems, welding, and circuit breakers. Here are some common instruments used:

1. Arc Detectors

- Optical sensors or photodiodes detect the light emitted by the electrical arc.

2. Current Sensors

- Current transformers or Rogowski coils measure the current flowing through the arc.

3. Voltage Sensors

- Voltage transformers or high-voltage probes measure the voltage across the arc.

4. Spectrometers

- Analyze the spectral characteristics of the arc, providing information on temperature, composition, and other properties.

5. High-Speed Cameras

- Capture the dynamics of the electrical arc, allowing for detailed analysis of arc behavior.

6. Data Acquisition Systems

- Record and process data from various sensors, enabling detailed analysis and monitoring of the electrical arc.

These instruments help researchers and engineers understand and optimize electrical arc phenomena in various applications, improving safety, efficiency, and performance.

Part(b)

Explain half wave potential with a suitable example.

ANS,,

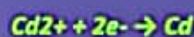
Half-wave potential ($E_{1/2}$) is a fundamental concept in electrochemistry, particularly in voltammetry.

Definition

Half-wave potential is the potential at which the current in a voltammetric experiment reaches half of its limiting value. It's a characteristic property of a redox couple and can be used to identify the species being oxidized or reduced.

Example

Consider the reduction of cadmium ions (Cd^{2+}) to cadmium metal:



In a voltammetric experiment, the current-potential curve would show a wave-shaped response. The half-wave potential ($E_{1/2}$) would be the potential at which the current reaches half of its limiting value.

Significance

The half-wave potential is significant because it:

1. ***Identifies the redox couple*:** $E_{1/2}$ is characteristic of a specific redox couple, allowing for identification of the species being oxidized or reduced.
2. ***Provides insight into redox properties*:** $E_{1/2}$ can provide information about the thermodynamics and kinetics of the redox reaction.

Factors Affecting Half-Wave Potential

1. ***Standard reduction potential*:** $E_{1/2}$ is related to the standard reduction potential of the redox couple.
2. ***Concentration*:** Changes in concentration can affect $E_{1/2}$.
3. ***Electrode material*:** The electrode material can influence $E_{1/2}$.